1. Clinical Question & Core Hypothesis

Clinical Question

Why do some TIL■rich bladder tumors fail to respond to PD■1/PD■L1 blockade despite apparent immune infiltration? 2) Tier 1 foundation: I built on the MSI/dMMR axis (Core Path 1) as the mutational/neoantigen driver of immunogenicity and on MSH6 (Core Path 2) as a specific MMR component whose loss can generate MSI/dMMR; I used Immunoscore concepts (Core Path 3) to justify using an H&E/WSI imaging surrogate of immune contexture. 3) Tier 2 support: TMB provides a quantitative neoantigen surrogate that can explain variation within MSI categories; liquid biopsy (ctDNA/CTCs) and urine DNA methylation offer noninvasive monitoring to test dynamic changes in tumor burden/response; anti■PD■1–linked tissue TIL dynamics provide mechanistic plausibility for re■invigoration of preexisting T cells. 4) Endpoint choice: I selected therapy_response (PD■1/PD■L1 response proxy) because the clinical puzzle concerns why immune infiltration does or does not translate into checkpoint inhibitor benefit. 5) Deliberate exclusions: I excluded exploratory candidates (TGF■β, stool SDC2/TFPI2 methylation, pZVR) from the core hypothesis because they are cross■tissue or less directly connected to MMR–TIL mechanistic axis and would dilute the focused, testable question?

Working Hypothesis

Hypothesis 1: In bladder cancer, the predictive value of high tumor-infiltrating lymphocyte (TIL) density for response to PD-1/PD-L1 blockade is contingent on concurrent mismatch-repair deficiency with MSH6 loss (MSH6 LOF), which drives MSI/dMMR and elevates tumor neoantigen burden—particularly indel-derived neoantigens—that license effective TIL-mediated cytotoxicity. As a result, within the TIL-high cohort, tumors that are MSI/dMMR due to MSH6 LOF will exhibit higher objective response rates (ORR), longer duration of response (DOR), and improved progression-free and overall survival (PFS/OS) to PD-1 blockade compared with TIL-high, microsatellite-stable (MSS) tumors. Hypothesis 2: Among TIL-high bladder cancers, those that are MSS will show limited benefit from PD-1 blockade despite immune infiltration, due to a relatively lower neoantigen load and/or a more suppressive microenvironment; noninvasive biomarkers such as circulating tumor DNA (ctDNA) dynamics and urine DNA methylation will correlate with therapeutic response and can monitor early treatment effects. Together, these imply that TMB, ctDNA clearance, and urine methylation changes will align with clinical benefit primarily in the TIL-high MSI/dMMR (MSH6 LOF) subgroup, validating an integrated biomarker strategy for patient selection and dynamic monitoring.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MSH6, MSI/dMMR
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6 (Core Path 2) as a specific MMR component whose loss can generate MSI/dMMR; I used Immunoscore concepts (Core Path 3) to justify using an H&E/WSI imaging surrogate of immune contexture.

MSI/dMMR: MSI/dMMR axis (Core Path 1) as the mutational/neoantigen driver of immunogenicity and on MSH6 (Core Path 2) as a specific MMR component whose loss can generate MSI/dMMR; I used Immunoscore concepts (Core Path 3) to justify using an H&E/WSI imaging surrogate of immune contexture.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	TMB	Supporting modifier from knowledge graph
Secondary	ctDNA/CTCs	Supporting modifier from knowledge graph
Secondary	Urine_DNA_methylation	Supporting modifier from knowledge graph
Secondary	Immunoscore (CD3/CD8 densities)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: The interaction between the imaging feature and anchor gene predicts clinical outcome.

Prediction 2: Spatial analysis will confirm the mechanistic link at the tissue level.

Prediction 3: Functional perturbation will establish causality.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- TMB: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- ctDNA/CTCs: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.